

Compressed Traffic Assignment with the Augmented Lagrangian Method*

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Abstract

We consider large-scale traffic assignment problems and develop an integrated major-minor path-compression and solution framework. Paths are partitioned according to nominal flows and a prescribed threshold: major paths are retained explicitly, while deviations of the minor-path flows from their nominal values are represented in a truncated singular-vector subspace of the minor path-link incidence matrix. A simple safeguard preserves feasibility, and the resulting lower-dimensional convex formulation is solved by an augmented Lagrangian method with separate penalty parameters for the equality and reconstructed nonnegativity constraints. The contribution is this integrated compression-and-solution framework; neither singular value decomposition nor the augmented Lagrangian principle is proposed in isolation.

We report computational studies using the Chicago Sketch, Chicago Regional, and Philadelphia networks. The revised experiments use a consistently converged uncompressed formulation as the primary reference and report preprocessing separately from the online augmented Lagrangian solution time. The results characterize a dimension-accuracy-computation trade-off: moderate compression can reduce the number of optimization variables and the cost per inner iteration while maintaining high link-flow fidelity, but total-time gains are problem-dependent because the reconstructed minor-path nonnegativity constraints retain path-scale operations. The framework is therefore intended for related or repeated assignment calculations in which a path pool and nominal flow information are already available and the one-time compression effort can be reused.

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1 Introduction

The traffic assignment problem is a classical problem in transportation science and network optimization. In its standard user-equilibrium formulation, it can be posed as a convex optimization problem that distributes travel demand across network paths subject to flow conservation and nonnegativity constraints. This formulation goes back to the seminal work of [BMW56], and it has motivated a large literature on both modeling and computation. In particular, the Frank-Wolfe method [FW56] became a standard approach because it avoids exhaustive path enumeration by solving shortest-path subproblems iteratively. Subsequent developments, including gradient projection, quasi-Newton, and conjugate-direction methods, have improved computational performance substantially; see, for example, [JTPR94], [Ber80, Ber11], [ML13], and the textbook treatments in [She85, AMO93, Pat94, Ber98, BLU20].

Despite this progress, path-based formulations remain difficult to solve at large scale because the number of feasible paths can be enormous. As network size grows, the associated path-link incidence matrix becomes very large, and the resulting optimization problem becomes increasingly expensive in both storage and computation. Existing methods alleviate this difficulty in different ways, for example, by generating paths selectively, refining active path sets, or exploiting origin-based structure. However, they do not directly reduce the dimension of the underlying path space. This dimensionality challenge is the starting point of the present work.

Our approach is motivated by a simple structural observation: in many realistic traffic assignment problems, *only a relatively small subset of paths carries substantial flow, while many other paths make smaller contributions to the aggregate link flows*. This suggests that the full path space may contain substantial redundancy from the standpoint of optimization. We exploit this structure by partitioning the paths into two groups. The first group consists of *major paths*, which are retained explicitly as decision variables. The second group consists of *minor paths*, whose contribution is represented approximately in a low-dimensional subspace obtained from a truncated singular value decomposition (SVD) of the corresponding minor-path submatrix of the path-link incidence matrix.

The proposed framework is aimed primarily at applications where closely related large-scale traffic assignment problems must be solved repeatedly. In such cases, nominal path-flow information from a previous solution, a nearby scenario, or a coarse preliminary calculation can be used to identify the major paths that should be retained explicitly. This information need not be exact; its role is only to guide the major-minor partition. Thus, our framework is not intended to replace a one-time full solve from scratch, but to exploit reusable nominal information in order to accelerate the solution of related instances.

The path pool is treated as an input to the compressed formulation. Accordingly, the paper does not propose a replacement for path generation or claim that compression is preferable for every traffic assignment calculation; it studies whether an already available path-based formulation can be reduced before repeated optimization.

Our construction yields a compressed path-based formulation with substantially fewer variables than the original problem. At the same time, the compressed problem preserves the basic convex structure of the original formulation. We also incorporate a simple safeguard that ensures feasibility of the compressed formulation whenever the original problem is feasible. Thus, the proposed method is more than a heuristic compression scheme: it yields a well-defined convex optimization problem that preserves the essential structure of the original traffic assignment model.

To solve the compressed problem, we use an augmented Lagrangian (AL) method with separate penalty parameters for the equality and reconstructed nonnegativity constraints. The paper is organized around one contribution: a nominal-flow-informed major–minor compression framework whose affine SVD representation and AL solution procedure are two inseparable components of the same construction. The AL principle is standard in isolation; its role here is to provide a reliable solution mechanism for the specific compressed formulation.

The remainder of the paper is organized as follows. In Section 2, we review related work on traffic assignment algorithms, spectral methods in transportation, and reduced-order convex programming. In Section 3, we develop the compressed formulation and establish its feasibility and approximation properties. In Section 4, we describe the AL method and its algorithmic properties. In Section 5, we report our computational studies on three networks. Section 6 concludes and discusses broader implications.

2 Background and Related Work

In this section, we review three strands of literature most relevant to our work. We first discuss path-based traffic assignment algorithms and their computational limitations at large scale. We then examine spectral and rank-based ideas in transportation, emphasizing the distinction between algebraic rank and the optimization-oriented use of truncated SVD in our framework. Finally, we discuss the connection between our approach and the broader literature on reduced-order convex optimization.

2.1 Traffic Assignment Problem and Solution Algorithms

Path-based formulations of the traffic assignment problem have been studied extensively, but their central computational difficulty has remained the same: the number of feasible paths can be extremely large, and the associated path–link incidence matrix grows rapidly with network size. This creates substantial challenges in both storage and computation for realistic transportation networks.

A classical way to address this difficulty is to avoid full path enumeration. The Frank-Wolfe algorithm and its variants generate shortest paths iteratively and have long served as standard methods for solving traffic assignment problems. Other path-based approaches,

including gradient projection, conjugate-gradient-type methods, and Newton-type methods, improve computational performance by refining active path sets and exploiting the smooth structure of the objective. These methods have substantially advanced the computational treatment of traffic assignment, but they still operate in the original high-dimensional path space.

Another line of work improves tractability by exploiting network structure. Origin-based and bush-based methods, have originated in both data and transportation network research, and organize computations around origins or acyclic subnetworks [Gal77, BGG84, BG02, Nie10]. Other approaches use greedy path management, decomposition, or operator-splitting techniques such as ADMM to improve scalability [XNL18, YZD⁺19, LCH⁺23, LDZ⁺24, LSX⁺25]. Mesoscopic and cross-resolution modeling frameworks [ZT14, ZCW⁺22], together with open-source tools such as `Path4GMNS` [LZ25b] and `AequilibraE` [Cam15], have further improved the practical study of large-scale traffic assignment problems.

However, these approaches primarily seek to manage or decompose the original path space rather than reduce its dimension directly. By contrast, our focus is on constructing a compressed path-based formulation in which the dimension of the decision space itself is reduced. In this sense, our approach is complementary to existing traffic assignment algorithms: rather than proposing another method for the full path representation, we reformulate that representation so that the resulting optimization problem is smaller.

2.2 Spectral Methods and Rank Structure in Transportation

Spectral and low-rank ideas have appeared in transportation in several different contexts. Early work used SVD and principal component analysis to support dynamic traffic assignment and online calibration [PB99, PSA⁺17]. Other studies examined the rank of the path-link incidence matrix from the standpoint of observability, sensor placement, and network reconstruction [HPC09, ZL10]. In these works, rank is interpreted in the algebraic sense, namely, as the number of linearly independent rows or columns needed to represent the matrix exactly.

This algebraic interpretation also relates to methods such as column generation and Dantzig-Wolfe decomposition, where linearly independent columns are generated so that the original constraint system is satisfied exactly in the limit, as the number of generated columns increases [LD05, BJN⁺98]. Our use of spectral structure is different. We do not use SVD to identify an exact basis for the path space, nor do we rely on rapid singular-value decay to justify compression from a matrix-approximation viewpoint alone. Instead, we use truncated SVD as a device for constructing a lower-dimensional representation of the contribution of minor paths to aggregate link flows.

This distinction is important for the present paper. In our experiments, the singular values of the minor-path incidence matrix decay gradually rather than rapidly. Thus, the matrix is not strongly low-rank in the usual spectral reconstruction sense. Nevertheless,

moderate-rank compression still performs well in the optimization problem: it preserves high link-flow fidelity and yields useful reductions in computational cost. This indicates that the effectiveness of the compression is governed less by accurate reconstruction of the minor-path matrix itself than by the relative insensitivity of the optimization problem to higher-order components of the minor-path representation.

Most recently, [ZQ25] constructed reduced hypothetical path sets with real-valued (non-binary) path-link incidence matrices by fitting a surrogate Logit-based stochastic user equilibrium model. Their approach provides evidence of low-dimensional structure in path-based traffic models, but it differs from ours in both objective and construction: they replace the original assignment model with a surrogate one, whereas we compress the original binary incidence matrix directly and solve a reduced formulation of the original Beckmann problem.

2.3 Reduced-Order Convex Programming

Low-rank reduction is also a common idea in large-scale optimization. In many problems, a high-dimensional model admits an effective lower-dimensional representation that preserves the components most relevant to the optimization objective while reducing computational cost. Such ideas appear in areas such as portfolio optimization [Mar52, Ros76, CZ93], machine learning [Vap95, SSM98, RR08], matrix approximation [EY36, GK65, SK87, SKKR00, KBV09, CR09], and aggregation in approximate dynamic programming [Ber12, Ber19].

Our work brings this reduced-order viewpoint to path-based traffic assignment. The key idea is to introduce low-rank representation not in observed traffic data, but in the path-based incidence matrix that defines the optimization problem itself. By projecting the minor-path component onto a low-dimensional spectral subspace, we obtain a reduced-order convex program that remains closely connected to the original traffic assignment model while involving substantially fewer decision variables.

This perspective places our work at the intersection of path-based traffic assignment and reduced-order convex optimization. The resulting formulation preserves convexity and, under a natural structural condition, feasibility as well. It can then be solved effectively by an AL method designed for the compressed structure.

3 Problem Formulation and Path-Based Compression

In this section, we describe our path-based compression method. We start by introducing the classical traffic assignment problem. We then introduce our compression framework, and provide a condition under which the compressed problem remains feasible.

3.1 Traffic Assignment Problem

We consider a transportation network with m directed links, n paths, and ℓ origin-destination (OD) pairs. Let A be the ℓ by n OD-path incidence matrix (rows correspond to OD pairs, and columns correspond to paths, with entries being either 1 or 0, depending on whether the column path corresponds to the row OD pair, or not). Let B be the n by m path-link incidence matrix (rows correspond to paths, and columns correspond to links, with entries being either 1 or 0, depending on whether the row path contains the column link, or not). There is a given nonnegative OD pair demand vector $d \in \mathbb{R}^\ell$. There is also a given non-negative offset link flow vector v_0 . In particular, the vector v_0 can represent the known contributions to the link flows by the OD pairs that have a unique path from origin to destination.

The traffic assignment problem is defined as

$$\begin{aligned} \min_{x,v} \quad & f(v) \\ \text{s.t.} \quad & v = B'x + v_0, \quad Ax = d, \quad x \geq 0, \end{aligned} \tag{1}$$

where f is a given convex function of the link flow vector v [specifically, the Bureau of Public Roads (BPR) congestion cost in our numerical experiments], a prime denotes transposition, and the inequality $\geq$ is meant to be component-wise.

Implicit in our formulation is that each OD pair has at least one path and that each path belongs to exactly one OD pair, so the problem is feasible, the matrix A has rank ℓ , and each column of A is a column of the $\ell \times \ell$ identity matrix. Since the cost function f is convex and the constraints are linear, problem (1) is a convex programming problem.

Despite being convex and feasible, problem (1) can be challenging to solve. This is because the values of n and ℓ are often quite large for practical problems. We will next introduce a compressed problem, which involves fewer variables, thus facilitating its computational solution.

3.2 Path Compression and Compressed Problem

In many practical contexts, nominal path-flow information is often available from historical data, prior computations, or coarse preliminary solutions, and can be used to identify the paths that are likely to be dominant. These are the *major paths* that have significant impact on the link flows v . In contrast, the remaining paths can be regarded as *minor paths*, whose contributions to the link flows are secondary. Our approach is to approximate the contributions of the minor paths to the link flows via SVD, as we discuss next.

In particular, let τ be a threshold on the nominal flow. For OD pairs that have multiple corresponding paths, we select major paths according to the following two rules:

- 1) For each OD pair, the path that has the largest nominal flow is a major path;

- 2) The remaining paths are split into major paths (those with nominal flow bigger than τ) and minor paths (those with nominal flow no more than τ).

The first rule ensures that the OD conservation constraints remain solvable under compression. We will discuss this in Section 3.3. The second rule allows us to use our knowledge of a nominal flow to simplify the problem.

Without loss of generality, let us assume that the first s elements of the vector x correspond to the major paths. As a result, we can partition the vector x as

$$x = \begin{pmatrix} y \\ w \end{pmatrix},$$

where $y \in \mathbb{R}^s$ and $w \in \mathbb{R}^{n-s}$ represent the flows of the major and minor paths, respectively. Accordingly, we can partition the matrix B as

$$B = \begin{pmatrix} B_1 \\ B_2 \end{pmatrix},$$

where the dimensions of B_1 and B_2 are s by m and $(n-s)$ by m , respectively. Suppose that B_2 is approximated via SVD as

$$B_2 \approx U_r \Sigma_r V_r',$$

where U_r is an $(n-s)$ by r matrix, and the dimensions of Σ_r and V_r are determined accordingly. The value r is a design parameter, which represents the extent to which the matrix B_2 is approximated.

By the Eckart–Young theorem, this truncated SVD is the best rank- r approximation of B_2 in the spectral norm; see [GVL13, Theorem 2.4.8]. Let the nominal minor-path flow be w_0 . Then the link-flow approximation error associated with a deviation from the nominal flow satisfies

$$\|(B_2 - U_r \Sigma_r V_r')(w - w_0)\| \leq \sigma_{r+1} \|w - w_0\|,$$

where $\|\cdot\|$ denotes the Euclidean norm and σ_{r+1} is the largest discarded singular value. We therefore approximate the minor-path flow by the affine representation

$$x = \begin{pmatrix} y \\ w_0 + U_r z \end{pmatrix},$$

where $z \in \mathbb{R}^r$. Thus, the nominal link-flow contribution $B_2' w_0$ is retained exactly, while the deviation $w - w_0$ is represented in the truncated singular-vector subspace. The matrix being decomposed remains B_2 ; only the minor-path flow representation is centered. The corresponding link flow is

$$\begin{aligned} v &= B'x + v_0 = \begin{pmatrix} B_1' & B_2' \end{pmatrix} \begin{pmatrix} y \\ w_0 + U_r z \end{pmatrix} + v_0 \\ &= B_1' y + D z + B_2' w_0 + v_0, \end{aligned}$$

where $D = B'_2 U_r$ is an m by r matrix.

Similarly, partition $A = (A_1 \ A_2)$, where A_1 is an ℓ by s matrix and A_2 is an ℓ by $(n - s)$ matrix. Then

$$Ax = A_1 y + Mz + A_2 w_0,$$

where $M = A_2 U_r$, and nonnegativity becomes

$$y \geq 0, \quad w_0 + U_r z \geq 0.$$

Writing $\bar{d} = d - A_2 w_0$, the compressed problem is

$$\begin{aligned} \min_{y,z} \quad & \hat{f}(y, z) \\ \text{s.t.} \quad & y \geq 0, \quad A_1 y + Mz = \bar{d}, \quad w_0 + U_r z \geq 0, \end{aligned} \tag{2}$$

where $\hat{f}(y, z) = f(B'_1 y + Dz + B'_2 w_0 + v_0)$. Setting $w_0 = 0$ recovers the uncentered formulation in the submitted version. We summarize the main notation in Table 9 given in Appendix A.

3.3 Convexity and Feasibility of the Compressed Problem

As discussed earlier, the original problem is convex and feasible. Clearly, we would like the compressed problem to remain convex and feasible. The convexity of the compressed problem follows directly from f being convex and the constraints in (2) being linear; see e.g., [Ber09, Prop. 1.1.4].

To see the feasibility of the affine compressed problem, suppose first that the nominal flow satisfies the OD demand constraints, so that there is a nominal major-path flow $y_0 \geq 0$ with $A_1 y_0 + A_2 w_0 = d$. Then $\bar{d} = A_1 y_0 \geq 0$, and $(y, z) = (y_0, 0)$ is feasible. If the nominal flow comes from a different demand, the nominal minor flows of any violating OD pair are proportionally reduced so that their total does not exceed that OD demand. This yields $\bar{d} \geq 0$. Because at least one major path is retained for every OD pair, $A_1 y = \bar{d}$ can then be satisfied by assigning $\bar{d}_i$ to one retained major path of OD pair i and setting its remaining major-path flows to zero. Thus, a feasible point with $z = 0$ always exists after this safeguard.

We note that the classical Beckmann formulation often involves a cost function that is strictly increasing with the link flows. As a result, the optimal link flows are unique, but the optimal path flows are generally not. This non-uniqueness property is inherent to the problem structure and is neither introduced nor resolved by the compression. Since solution quality is determined by the link flows rather than the path flows, the compressed formulation can achieve high fidelity without recovering individual minor-path flows.

Figure 1 summarizes the six steps of our compression procedure, from the original problem with $n + m$ variables to the reduced formulation with $s + r$ variables.

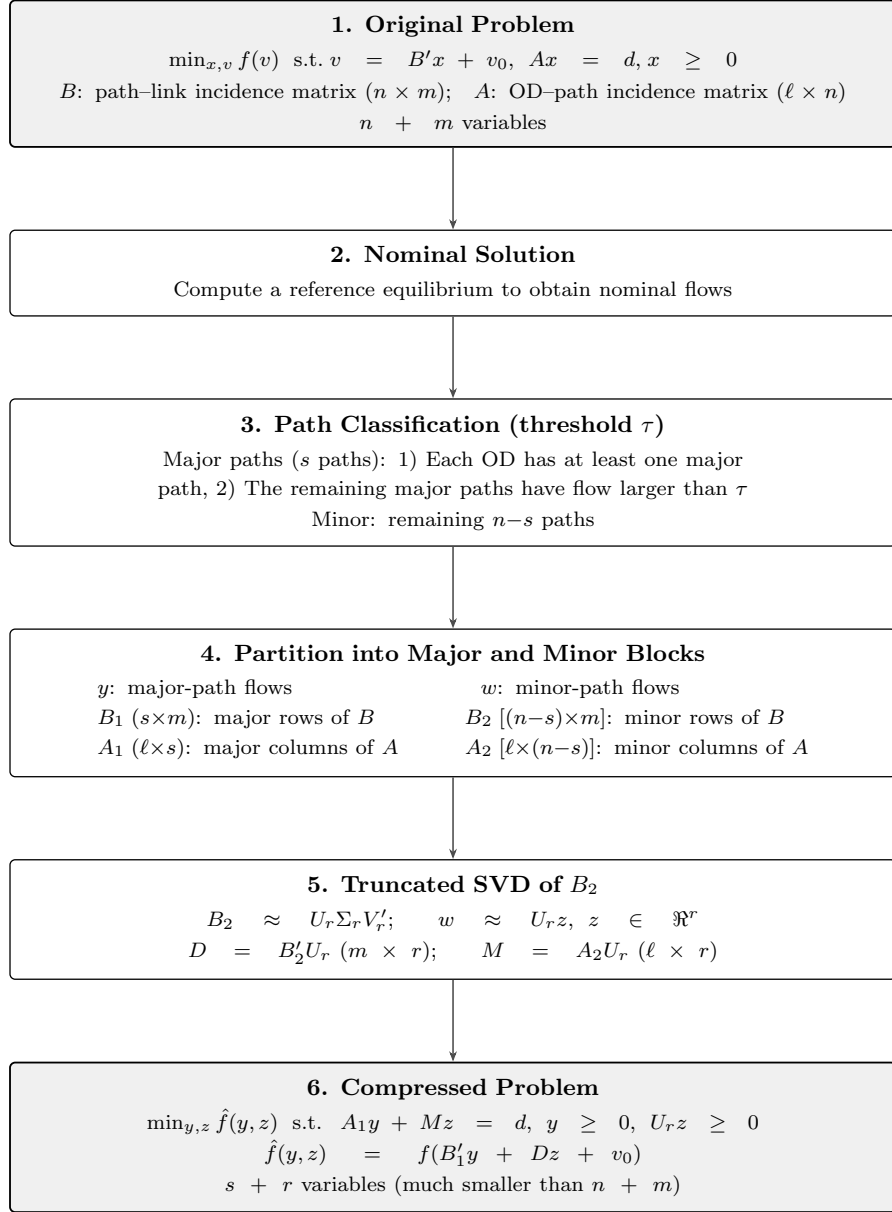


Figure 1: Architecture of the major-minor path compression framework. A nominal solution partitions paths into major (retained) and minor (compressed via truncated SVD of B_2), yielding a formulation with $s + r$ variables, which can be a lot less than the $n + m$ variables of the original problem.

Revision note: In the final clean figure, replace $w \approx U_r z$ by $w \approx w_0 + U_r z$ and update the Step 6 equality, nonnegativity constraint, and objective consistently. No other architectural change to the figure is proposed.

4 Solution via the Augmented Lagrangian Method

We solve problem (2) via the AL method with partial elimination of constraints; see, e.g., [Ber82, Section 2.4], [Ber16, Section 5.2]. There are two reasons for this choice. First, the AL method is known to be reliable and enables the use of efficient computational tools for optimization problems with simple inequality constraints, such as $x \geq 0$ and $y \geq 0$. Second, solutions from previous iterations in the AL method can be used to facilitate the computation of subsequent iterations. These features make the AL method well-suited for addressing the compressed problem (2). In what follows, we describe the exact AL method that we use, including the AL function and multiplier update formulas for each iteration.

4.1 Augmented Lagrangian Function

We denote by λ and μ the multipliers associated with the constraints $A_1 y + Mz - \bar{d} = 0$ and $w_0 + U_r z \geq 0$, respectively. As a result, we have $\lambda \in \mathbb{R}^\ell$ and $\mu \in \mathbb{R}^{n-s}$. For these two sets of constraints, we also associate different penalty parameters, which define a two-dimensional vector c , i.e., $c = (c_1, c_2)$, where c_1 and c_2 are associated with the constraints $A_1 y + Mz - \bar{d} = 0$ and $w_0 + U_r z \geq 0$, respectively. We will also use the subscript i to denote the i th component of a vector. In particular, $[w_0 + U_r z]_i$ is the i th component of the vector $w_0 + U_r z$.

Using the notation introduced above, the AL function is given as:

$$\begin{aligned} L_c(y, z, \lambda, \mu) &= \hat{f}(y, z) + \lambda'(A_1 y + Mz - \bar{d}) + \frac{c_1}{2} \|A_1 y + Mz - \bar{d}\|^2 + \\ &\quad \frac{1}{2c_2} \sum_{i=1}^{n-s} \left\{ \left(\max\{0, \mu_i - c_2[w_0 + U_r z]_i\} \right)^2 - \mu_i^2 \right\}. \end{aligned} \quad (3)$$

The function $L_c(y, z, \lambda, \mu)$ is continuously differentiable with respect to y and z . The gradients are given by:

$$\begin{aligned} \nabla_y L_c(y, z, \lambda, \mu) &= \nabla_y \hat{f}(y, z) + A_1' (\lambda + c_1(A_1 y + Mz - \bar{d})), \\ \nabla_z L_c(y, z, \lambda, \mu) &= \nabla_z \hat{f}(y, z) + M' (\lambda + c_1(A_1 y + Mz - \bar{d})) - U_r' (\mu + c_2 h^+(z, \mu, c_2)), \end{aligned}$$

where

$$h^+(z, \mu, c_2) = \begin{pmatrix} h_1^+(z, \mu, c_2) \\ h_2^+(z, \mu, c_2) \\ \vdots \\ h_{n-s}^+(z, \mu, c_2) \end{pmatrix}, \quad h_i^+(z, \mu, c_2) = \max \left\{ -[w_0 + U_r z]_i, -\frac{\mu_i}{c_2} \right\}. \quad (4)$$

A detailed derivation of the preceding formulas for problems with general inequality constraints can be found in several textbooks, see e.g., [Ber82, Chapter 3].

4.2 Iterations of the Augmented Lagrangian Method

Let us discuss the updating equations for the multipliers and penalty parameters. Given the multipliers of the k th iteration, which we denote as λ^k and μ^k , and the penalty parameter $c^k = (c_1^k, c_2^k)$, we first compute the values y^k and z^k via the following minimization:

$$(y^k, z^k) \in \arg \min_{y \geq 0, z} L_{c^k}(y, z, \lambda^k, \mu^k). \quad (5)$$

In our implementation, this inner minimization is solved using the L-BFGS-B code, which handles the simple bound constraint $y \geq 0$ by using the gradient projection method developed by the last author in [Ber82, Section 1.5].

Having computed the values y^k and z^k , the updated multipliers λ^{k+1} and μ^{k+1} are given by:

$$\begin{aligned} \lambda^{k+1} &= \lambda^k + c_1^k (A_1 y^k + M z^k - \bar{d}), \\ \mu^{k+1} &= \mu^k + c_2^k h^+(z^k, \mu^k, c_2^k), \end{aligned} \quad (6)$$

cf. Eq. (4).

To accelerate convergence while controlling ill-conditioning in the AL minimization, we update separately the penalty parameters associated with different constraints. In particular, we fix some parameters $\beta > 1$ and $\gamma \in (0, 1)$, and set $c^{k+1} = (c_1^{k+1}, c_2^{k+1})$ according to:

$$\begin{aligned} c_1^{k+1} &= \begin{cases} \beta c_1^k & \text{if } \|A_1 y^k + M z^k - \bar{d}\|_\infty > \gamma \|A_1 y^{k-1} + M z^{k-1} - \bar{d}\|_\infty, \\ c_1^k & \text{otherwise;} \end{cases} \\ c_2^{k+1} &= \begin{cases} \beta c_2^k & \text{if } \|h^+(z^k, \mu^k, c_2^k)\|_\infty > \gamma \|h^+(z^{k-1}, \mu^{k-1}, c_2^{k-1})\|_\infty, \\ c_2^k & \text{otherwise,} \end{cases} \end{aligned} \quad (7)$$

where $\|\cdot\|_\infty$ denotes the infinity norm.

The convexity of our problem guarantees convergence of our AL method under our conditions. We refer to [Ber82], Chapter 5, for a general discussion of the convergence and rate of convergence issues of the AL method for convex constrained optimization. Generally, the convergence rate of an AL method depends on the choice of the cost function f as well as the details of the termination criterion (7), which are usually tuned by experimentation.

4.3 The Compression Mapping and Its Solution

Let Φ_r be a fixed $(n - s) \times r$ matrix. The path-flow vector is represented as

$$x = \begin{pmatrix} y \\ w_0 + \Phi_r z \end{pmatrix}, \quad \Phi_r = U_r \quad \text{for the truncated-SVD representation used in this paper.} \quad (8)$$

The columns of Φ_r are the retained minor-flow directions and the components of z give their coefficients relative to the nominal flow w_0 . The same mapping admits a simple profile interpretation. If a column of Φ_r is a prescribed nonnegative path-selection profile, its corresponding component of z is the total flow assigned to that profile. This observation is only a guiding interpretation; the analysis and computations use $\Phi_r = U_r$.

Substitution in the original problem gives

$$\begin{aligned} \min_{y,z} \quad & \hat{f}_\Phi(y, z) = f(B'_1 y + B'_2(w_0 + \Phi_r z) + v_0) \\ \text{s.t.} \quad & (y, z) \in \mathcal{C}_\Phi, \end{aligned} \quad \mathcal{C}_\Phi = \left\{ (y, z) \mid \begin{array}{l} A_1 y + A_2(w_0 + \Phi_r z) = d, \\ y \geq 0, \quad w_0 + \Phi_r z \geq 0 \end{array} \right\}. \quad (9)$$

For every fixed Φ_r , problem (9) is convex. If $w_0 \geq 0$ and $A_1 y_0 + A_2 w_0 = d$ for some $y_0 \geq 0$, then $(y_0, 0)$ is feasible. Thus, the SVD is not required for convexity; its role is to provide the particular map $\Phi_r = U_r$ and the approximation property established in Section 3.2.

The mapping and the solution method have different roles. The augmented Lagrangian method of Sections 4.1–4.2 is used in this paper. For illustration, a gradient-projection method for the same compressed convex problem is

$$\xi^{k+1} = P_{\mathcal{C}_\Phi} \left[\xi^k - \alpha_k \nabla \hat{f}_\Phi(\xi^k) \right], \quad \xi = (y, z). \quad (10)$$

A Frank–Wolfe method can be defined by replacing the projection in Eq. (10) with a linear minimization over $\mathcal{C}_\Phi$. These methods have different iterates and computational requirements, but they solve the same fixed compressed formulation. Consequently, the dimension reduction summarized in Table 1 is a property of the representation and can benefit the AL method or another suitable solution method; it is not produced by the augmented Lagrangian principle alone.

The reduction concerns the number of independent optimization variables. The reconstructed constraint $w_0 + U_r z \geq 0$ still has $n - s$ components and U_r is generally dense. This does not, however, imply that the dense matrix must dominate the computation. The factored calculation $B'_2 U_r z \approx V_r \Sigma_r z$ avoids forming the dense matrix $B'_2 U_r$, which reduces but does not remove the dense work; Section 5.2 makes the resulting trade-off explicit. The numerical study therefore reports both the dimension ratio and the realized speedup rather than inferring speed solely from sparsity.

5 Computational Studies

The computational study answers one question: *when does compressing the minor-path variables actually reduce solution time, and by how much?* The submitted version reported threshold and rank sweeps on three road networks. Re-running those experiments exposed a sharper structure than the sweeps alone revealed, and the study is organized around it here: a cost model that predicts when compression pays (§5.2); a controlled grid family

Table 1: Full and compressed path-flow representations. Here $K = n/\ell$ is the average number of candidate paths per multi-path OD pair and $q = (s + r)/\ell$ is the effective number of represented variables per OD pair.

Representation	Variables	Dimension	Mapping and compression meaning
Full path flow	x	$n = \ell K$	$v = B'x + v_0$; every candidate path is an independent coordinate.
Compressed	(y, z)	$s + r = \ell q$	$x = (y, w_0 + \Phi_r z)$; the minor-path block is represented by r retained directions.
Compression factor		$\rho = \frac{n}{s + r} = \frac{K}{q}$	As K/q increases, the same reduced coordinates represent a richer candidate path pool. The reduction is available to ALM and to another method applied to the same compressed feasible set.

that isolates the two variables the model identifies (§5.3); the rank, which turns out to be a speed parameter rather than an accuracy parameter (§5.4); the criterion selecting the retained subspace, where weighting by the nominal flow halves the objective gap on route-rich instances at no additional cost (§5.5); the affine offset w_0 , without which the representation does not work (§5.6); the solution operator, where the benefit turns out to be specific to the augmented Lagrangian (§5.7); and the road networks, where the model’s predictions are tested against practice (§5.8). Section 5.10 states the scope and the negative results.

[Note to co-authors: This section is a full rewrite rather than an update of the submitted one, so it is set in black and red is reserved for these notes. The organizing change is that it leads with a cost model and then tests it, instead of reporting parameter sweeps. Referee 1’s own diagnosis — that the dense $(n - s) \times r$ nonnegativity block caps the gain — is exactly what the model formalizes, which is deliberate.]

5.1 Experimental Setup

Instances. Two families are used. The *grid family* is an $N \times N$ Manhattan grid with bidirectional links, unit free-flow time, capacity 600, and BPR parameters $(\alpha, \beta) = (0.15, 4)$; demand runs from the left column to the right column with $720/N$ vehicles per OD pair, so the mid-cut volume-capacity ratio stays near 1.2 at every N and congestion does not drift as the grid grows. Candidate paths are generated by a deterministic penalty K -shortest-path routine at depth K . The *road networks* are the three instances of the submitted study, Chicago Sketch, Chicago Regional, and Philadelphia (Table 7), together with Sioux Falls and a sequence of progressively enriched Chicago Sketch pools used to vary route richness at fixed network and demand.

Bases. Every compressed instance is solved with both subspace criteria — the plain singular subspace of B_2 and the flow-weighted subspace of $\text{diag}(\sqrt{w_0})B_2$ (Section 5.5) — run back to back, so each comparison is within-session and unaffected by machine load.

Implementations. All experiments are run in two independent implementations of the same algorithm: a Python/NumPy reference and a compiled C++ solver. Reported speedups are always *within-implementation* ratios — an implementation’s compressed time divided into its own uncompressed time on the same instance — because, as §5.10 documents, the ratio depends on the relative throughput of dense and sparse kernels and is not portable between implementations.

Protocol. Single thread, matched stopping tolerance 10^{-4} on a common relative duality-gap certificate, quiet machine. Every terminal iterate is converted to a feasible point by the same per-OD projection before any objective is compared; δ_F reports the ℓ_1 mass that conversion moves and Gap_F the resulting feasible objective gap against the best feasible point found by any method on that instance. Timings on the smaller instances are medians of three runs; on the largest instances they are single runs, and repeated identical solves varied by as much as 40% under machine load. It is worth stating this precision explicitly, because it governs which comparisons the study is entitled to make. Re-measuring identical solves — same instance, same settings, same binary — across the campaign gave spreads of 1.1 to 1.6 $\times$ typically and 2.8 $\times$ in the worst case, so we treat 1.45 $\times$ as a conservative noise floor and read no speed difference smaller than that as an effect. Feasible objective gaps carry no such caveat: they are deterministic and reproduced bit for bit across every repetition, so accuracy comparisons are exact even when the corresponding timings are not.

[Note to co-authors: Two accounting decisions are ours to make before submission. (i) None of the reported speedups charge the SVD or the nominal-flow computation; Referee 1 objected to exactly this and the objection is currently unanswered. Proposal: report every ratio twice, online-only and fully charged, with a break-even count in related solves. (ii) Large-instance timings are single runs at $\pm 40\%$; medians of three would cost roughly two machine-hours and would let the near-parity entries be read as ordered.]

5.2 When Compression Pays: A Cost Model

Compression replaces n path variables by $s+r$, but the gradient of the compressed problem is not simply $(s+r)/n$ times cheaper. Each evaluation must reconstruct the minor flows to enforce $w_0 + U_r z \geq 0$, and U_r is dense: the reconstruction and the corresponding gradient term each cost about $(n-s)r$ operations, so the compressed gradient pays approximately $2r$ dense operations *per minor path*. The uncompressed gradient, by contrast, pays the sparsity of a path — the number of links it contains — per path. The comparison is therefore not between n and $s+r$ but between

$$2r \quad \text{and} \quad (\text{average links per path}). \quad (11)$$

Compression reduces work per evaluation when $2r$ is at most the average path length, and adds work when it is not. Two consequences are tested in what follows. First, the benefit should grow with the size of the candidate pool at fixed network, because the uncompressed side grows with the pool while the compressed side does not. Second, the rank should behave mainly as a cost parameter, since it enters (11) linearly but affects the represented subspace only weakly once the major paths carry the bulk of the flow.

5.3 A Controlled Study on Grid Networks

The grid family isolates the two quantities in (11): the grid dimension N sets the average path length ($\approx N$ links), and the generation depth K sets the pool size. Table 2 reports the speedup of the compressed augmented Lagrangian over the uncompressed one at fixed threshold (75th percentile of the nominal flow, giving $\approx 74\%$ variable reduction) and rank $r = 20$, in the C++ implementation.

Table 2: Speedup of compressed over uncompressed ALM on the grid family (C++, $r = 20$, $\approx 74\%$ variable reduction, matched tolerance). Speedup rises with pool depth K at every grid size. Feasible objective gaps are at most 0.55% in the $K \geq 32$ columns and 1.4–3.4% in the $K = 4$ column, where the fixed threshold removes genuine flow from a thin pool.

	$K = 4$	$K = 8$	$K = 16$	$K = 32$	$K = 64$
$N = 8$	0.55	0.59	1.53	1.45	2.01
$N = 16$	2.09	0.89	1.75	5.16	6.82
$N = 24$	2.25	1.18	0.73	3.87	4.86
$N = 32$	1.76	2.76	0.96	3.13	5.02

The prediction holds (Figure [effig:speedup-mechanismsa](#)). Reading across any row, the speedup rises with pool depth, reaching $6.82\times$ at $N = 16$, $K = 64$ with a feasible objective gap of 0.0000%; the $K = 32$ and $K = 64$ columns are uniformly between $3.1\times$ and $6.8\times$ with gaps at most 0.55%. Reading down the columns, the effect strengthens with grid size once the pool is deep enough, consistent with longer paths making the right-hand side of (11) larger. The $K = 8$ –16 interior is noisy because those cells solve in under ten seconds. The $K = 4$ column carries 1.4–3.4% gaps: with only four candidate paths per OD pair, a fixed 75th-percentile threshold classifies genuinely used paths as minor.

[Note to co-authors: Open question for the group: the grid is where the largest speedups live, but it is synthetic, and our own verification found the C++ uncompressed baseline on grids converging to an objective 4.6–4.9% above the Python one. Until that baseline is tightened, a referee can argue these ratios compare two under-converged solves. Do we tighten the baseline, or present the grid strictly as a mechanism study and let the road networks carry the efficiency claim?]

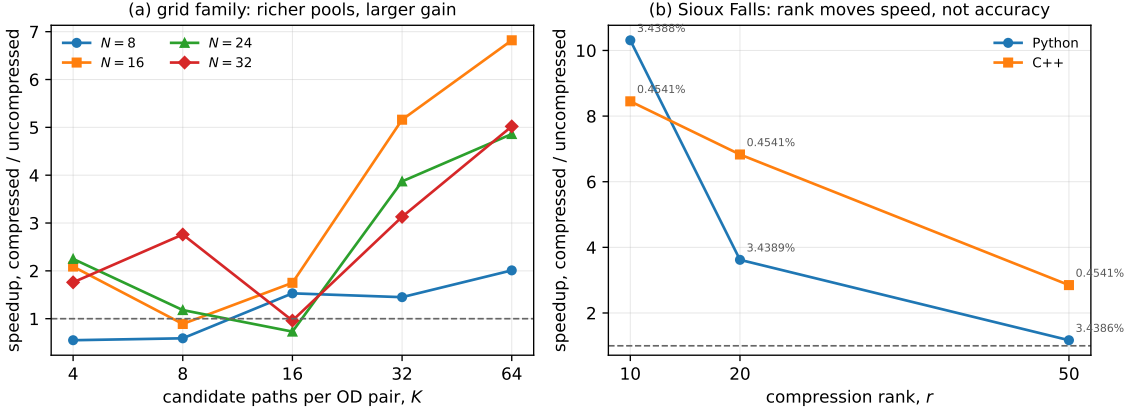


Figure 2: The two mechanisms that govern the compression speedup. (a) On the grid family, the gain grows with the depth of the candidate pool at every grid size, because the uncompressed problem grows with the pool while the compressed dimension does not; the dashed line marks parity. (b) On Sioux Falls the rank moves the speedup by close to an order of magnitude in both implementations while the feasible objective gap, annotated at each point, is unchanged to four decimal places — over this range the rank is a speed parameter, not an accuracy parameter.

5.4 The Rank is a Speed Parameter

Equation (11) predicts that the rank controls cost linearly. The submitted study swept $r \in \{50, 100, 150, 200\}$ and found solution quality insensitive to r while cost grew — correct, but sampled entirely on the expensive side. Table 3 sweeps downwards on Sioux Falls in both implementations.

The speedup varies by nearly an order of magnitude while the feasible objective gap moves in the fourth decimal place (Figure [effig:speedup-mechanismsb](#)). Over this range the rank is therefore almost purely a speed parameter: it multiplies the dense reconstruction without moving the solution. The optimum is interior rather than at $r \rightarrow 0$, because too small a subspace eventually lengthens the inner solves — visible on the grid, where $N = 16$, $K = 32$ gives $5.24\times$, $7.34\times$ and $5.44\times$ at $r = 10, 20$ and 40 . Combining this with (11) gives the design rule used throughout the rest of the study: *choose $2r$ no larger than the average path length*, which on these instances means $r \approx 10$ – 20 rather than the $r = 50$ of the submitted experiments. Section 5.8 quantifies what that choice is worth on a road network.

[Note to Yuchao: This rule is the direct answer to Referee 2’s request for analytical guidance on r , and it is currently the only place we give a parameter rule. He asked the same question about τ — uniform or per-OD, and how it relates to convergence — and we still choose τ by quantile with no argument. If you can supply even a heuristic derivation for τ analogous to (11), it would close his second

Table 3: Effect of the rank on Sioux Falls ($\tau = 600$, 91% variable reduction, matched tolerance, medians of three runs). Speedup is against each implementation’s own uncompressed baseline (1.34s Python, 8.20s C++). The feasible objective gap is unchanged to four decimal places across the entire range.

Implementation	r	CPU (s)	Speedup ($\times$)	Feasible obj. gap (%)
Python	10	0.13	10.31	3.4388
	20	0.37	3.62	3.4389
	50	1.15	1.17	3.4386
C++	10	0.97	8.45	0.4541
	20	1.20	6.83	0.4541
	50	2.88	2.85	0.4541

objection as cleanly as this closes the first.]

5.5 Which Subspace to Keep: Flow-Weighted Selection

Section 5.4 fixes how large the retained subspace should be. A separate question is which subspace of that size to retain. The natural answer, and the one used above, is the leading left-singular subspace of B_2 , which minimises $\|B_2 - \Pi B_2\|_F$. That criterion counts every minor path equally, and it is not the criterion the problem asks for: a minor path enters the link load in proportion to the flow it carries, so an error on a path carrying a thousand vehicles matters a thousand times more than the same error on a path carrying one.

Replacing B_2 by $\text{diag}(\sqrt{w_0}) B_2$ before the factorisation retains instead the subspace minimising $\sum_p w_{0,p} \| (B_2 - \Pi B_2)_p \|^2$, so the rank budget is spent where the flow is. The two factorisations are the same size and cost the same — 0.94s against 0.97s on Chicago Sketch at $r = 20$ — so any difference in solution time is a convergence effect and not preprocessing overhead. Table 4 varies the candidate pool at fixed network, demand, threshold and rank.

Table 4: Effect of flow-weighted subspace selection on Chicago Sketch ($\tau = 4.54$, $r = 20$, matched tolerance). Network, demand and parameters are fixed; only the richness of the candidate pool varies. Both bases are solved back to back on each instance.

$\bar{K}$	flow-weighted		plain		gap reduction
	speedup	gap (%)	speedup	gap (%)	
2.45	1.62	0.0284	1.55	0.0291	−2.4%
3.49	3.71	0.0202	3.34	0.0232	−12.9%
8.64	1.55	0.2086	1.34	0.7024	− 70.3%
11.09	0.83	0.4350	0.66	1.1914	− 63.5%
15.34	1.20	0.8649	1.15	1.8651	− 53.6%

The effect is on accuracy, and only on accuracy. The gap column improves at every richness point, by more than half wherever $\bar{K} \geq 8.64$, and these figures are exact: feasible objective gaps are deterministic and reproduce bit for bit across runs. The speedup column moves in the same direction, but by margins of 4 to 26%, which is inside the measurement noise reported in Section 5.1. Pooling all 34 controlled weighted-versus-plain pairs measured in this study — each a single instance with both bases solved back to back — the gap improves in 27, whereas the speedup improves by more than the noise floor in only 2. We therefore claim an accuracy improvement and explicitly do not claim a speed improvement.

The trend across the table is the explanation: a rich candidate pool is rich precisely in paths that carry almost no flow, and the unweighted criterion spends its rank budget representing them. This is why the regime that was least favourable in Section 5.8 — rich pools, where the compressed problem showed the largest gap — is the regime the weighting repairs most.

The grid family reproduces the effect on the axis that can be resolved. Across its twenty cells the weighted basis lowers the feasible gap in 15, taking the mean from 1.285% to 1.112%, with the largest improvements again at high pool depth (at $N = 32$, $K = 64$ the gap falls from 0.406% to 0.000%). Sioux Falls, at 91% variable reduction, shows the same sign and the largest relative movement, 11 to 20%. The conclusion is that weighting makes the subspace criterion consistent with the objective — the retained directions are the ones that carry flow — and that this buys accuracy at no preprocessing cost, not speed.

5.6 The Affine Offset is Load-Bearing

The representation is affine, $x_{\text{minor}} = w_0 + U_r z$, and the offset w_0 is the nominal minor flow. Because U_r is signed, the feasible set $\{w_0 + U_r z \geq 0\}$ contains the starting point $z = 0$ in its interior only by virtue of w_0 ; without the offset, $z = 0$ lies on the boundary of every reconstructed row simultaneously. To test whether this matters we remove the offset exactly — setting $w_0 = 0$ while moving its flow into the base link volume and the effective demand, so the model is unchanged — and re-solve.

The effect is large. Across the twenty grid cells of Table 2 the mean speedup falls from $2.60\times$ to $1.07\times$ and the mean feasible gap grows from 0.75% to 2.42%; the cells where compression is strongest collapse ($N = 24$, $K = 64$ from $4.86\times$ to $0.45\times$; $N = 16$, $K = 64$ from $6.82\times$ to $1.86\times$). On Sioux Falls the compressed solve slows from 1.7s to 7.9s, a fall from $6.2\times$ to $1.34\times$, with the cost per inner iteration rising fourfold at nearly unchanged iteration count: with the hinge active at the iterate, the augmented Lagrangian subproblem becomes nonsmooth exactly where the solver works. In the Python implementation the offset-free variant is not slower, and the accuracy penalty appears only on the larger instances, so the speed half of this conclusion is specific to the compiled solver while the accuracy half is not. The offset is therefore not an implementation convenience: *a signed subspace of nonnegative flows is usable only relative to an interior anchor*, and w_0

supplies it.

The two representational choices are independent. Removing the offset from the *flow-weighted* representation of Section 5.5 degrades it by the same mechanism and by a similar amount: across the grid family the mean feasible gap rises from 1.112% to 1.394%, and individual cells collapse in the same way (at $N = 32$, $K = 64$, $5.41\times$ at a gap of 0.000% becomes $1.00\times$ at 1.681%). Choosing the subspace better does not remove the need for an anchor inside it, and choosing the anchor well does not substitute for aligning the subspace with the flow. The two act on different parts of the representation and their benefits do not overlap.

[Note to co-authors: This result cuts both ways and we should decide how to present it. It explains why the representation works, but it also quantifies the dependence Referee 1 objected to: w_0 is obtained by solving the full problem, and we have now proved the method needs it. The strongest available answer is an experiment we have not run — reuse a *stale* w_0 from a perturbed instance and show the speedup survives. That single test would answer the circularity objection, the preprocessing objection and the “which application setting” objection at once. Recommend we run it before resubmission.]

5.7 Which Operator Benefits from Compression

The compressed representation is independent of the solution method, so one may ask whether the benefit is a property of the representation or of the augmented Lagrangian. We solve the same compressed instances with three operators: the augmented Lagrangian; gradient projection on the compressed set; and a reduced-gradient method that eliminates OD conservation by anchor substitution. Table 5 reports the result on grid cells in the regime where compression is effective.

Table 5: Speedup from compression, by solution operator, on grid cells (Python, w_0 representation, matched tolerance). Each column is the operator’s own uncompressed time divided by its compressed time. Frank–Wolfe has no compressed counterpart because the linear oracle over the compressed set is itself a linear program.

Grid cell	ALM	Gradient projection	Reduced gradient
$N = 16, K = 16$	1.11	0.00	0.07
$N = 16, K = 32$	1.22	0.01	0.26
$N = 16, K = 64$	1.96	0.01	0.13
$N = 32, K = 16$	3.85	0.02	0.34
$N = 32, K = 32$	1.74	0.04	0.22
$N = 32, K = 64$	2.54	0.10	0.26

Only the augmented Lagrangian benefits. The reason is structural rather than incidental. An iteration costs gradient work over the *variables* plus feasibility work over the *constraints*; compression reduces the former and, because the reconstruction is dense, increases the latter. The augmented Lagrangian carries feasibility in the objective through

multipliers and a penalty, so its iteration cost is gradient-dominated and the variable reduction passes through to wall-clock time. Gradient projection instead derives its speed from a projection that is separable across OD pairs and available in closed form; on the compressed set $\{A_1 y + Mz = d_{\text{eff}}, y \geq 0, w_0 + U_r z \geq 0\}$ the latent variables z appear in every OD equality and in every reconstructed bound, so separability is lost and the projection degrades into a coupled quadratic program. On the road networks this is not merely slower but infeasible: the exact projection requires a dense factorization of order the number of OD pairs, which at $\ell = 17,464$ does not scale. The reduced-gradient method removes the OD equalities by substitution, but the binding cost is the dense reconstruction, which substitution does not touch; what elimination removes, the representation restores.

We therefore state the scope explicitly: *variable reduction accelerates a method whose per-iteration bottleneck lies in the variables, not in the constraints*. This is the measured justification for pairing the compressed representation with an augmented Lagrangian rather than a projection-type method, and it is the reason the framework of Section 3 is built that way.

[Note to co-authors: Referee 2 wrote that “the augmented Lagrangian method is quite standard” and wanted a tailored algorithm. This subsection is our answer: not a new algorithm, but a demonstration that the pairing is forced. Whether that satisfies him is the main uncertainty in the revision. He also asked for comparison against state-of-the-art path-based methods; we compare four operators but no bush-based method (Algorithm B, OBA, iTAPAS). Do we add one, or argue the comparison is out of scope?]

5.8 Road Networks

Three questions remain for the road networks: do the operators agree on the same answer; does the richness prediction of §5.2 hold on a real network; and what does the rank rule of §5.4 buy in practice.

Consistency of the operators

Before comparing efficiency we verify that the methods share a numerical target. Table 6 solves one instance with all operators under matched tolerances and a common feasibility conversion.

The compressed rows agree with each other to within 0.03 percentage points of feasible objective gap, confirming that the gap is a property of the representation at this (τ, r) and not of the operator, while the uncompressed rows bracket the reference to within 0.05%. Frank–Wolfe is the most accurate uncompressed operator at loose tolerance and is used as an honest baseline where relevant.

Table 6: Numerical consistency on Sioux Falls under matched stopping tolerances. The reference is the best feasible point over the uncompressed operators (full-path gradient projection, feasible objective 4.231×10^6 , certificate 9.1×10^{-10}). The augmented Lagrangian’s feasibility-based stopping rule leaves it 0.048% above that reference; the near-agreement of all operators is the consistency this table quantifies.

Representation	Method	Raw obj. diff. (%)	$\ A\bar{x} - d\ _\infty$	δ_F (%)	Feasible obj. gap (%)	Link-flow diff. (%)
Full ($\tau = 0$)	ALM	0.048	8.7×10^{-11}	1.3×10^{-9}	0.048	1.156
Full ($\tau = 0$)	Frank-Wolfe	1.2×10^{-4}	2.4×10^{-10}	2.7×10^{-12}	1.2×10^{-4}	5.1×10^{-4}
Full ($\tau = 0$)	Gradient projection	9.8×10^{-8}	5.5×10^{-11}	1.5×10^{-11}	9.8×10^{-8}	0.005
Compressed ($\tau = 600, r = 20$)	ALM	3.489	1.7×10^{-10}	1.5×10^{-5}	3.489	4.679
Compressed ($\tau = 600, r = 20$)	Gradient projection (signed)	3.308	7.8×10^{-11}	0.102	3.461	4.655

Table 7: Network characteristics. Singleton paths correspond to the OD pairs with a unique feasible path and are excluded from the compressed problem.

	Chicago Sketch	Chicago Regional	Philadelphia
Nodes / Links	933 / 2,950	12,982 / 39,018	13,389 / 40,003
Multi-path variables (n)	42,774	879,625	1,846,068
Multi-path OD pairs (ℓ)	17,464	353,582	601,644
Average paths per multi-path OD ($\bar{K} = n/\ell$)	2.45	2.49	3.07
Multi-path flow share	23.5%	59.2%	70.9%

Route richness and the rank rule

Table 8 varies route richness on Chicago Sketch at fixed network and demand, by enriching the candidate pool from the submitted $\bar{K} = 2.45$ to $\bar{K} = 15.34$, and reports the same axis at the rank chosen by §5.4 and at the submitted rank.

Table 8: Route-richness axis on Chicago Sketch: same network and demand, candidate pools of increasing average richness $\bar{K}$, $\tau = 4.54$, Python implementation. The $r = 20$ column applies the rule of §5.4; the $r = 50$ column is the submitted rank, shown for comparison at identical accuracy.

$\bar{K}$	$\bar{K}/q$	Full dim. n	Compressed dim. $s + r$	Speedup ($r = 20$)	Speedup ($r = 50$)	Feasible obj. gap (%)
2.45	2.14	42,774	20,017	1.17	0.98	0.03
3.49	3.37	229,175	67,986	4.84	1.95	0.02
8.64	7.77	709,567	91,297	1.00	0.66	0.72
11.09	9.84	903,516	91,780	0.74	0.67	1.18
15.34	13.21	1,323,999	100,185	1.07	0.67	1.88

Two readings. First, the rank rule is worth a factor of about 2.5 at the peak — $4.84\times$ against $1.95\times$ — at accuracy identical to two decimal places, so a substantial part of the attainable speedup in the submitted study was lost to the rank choice alone. Second, and

unlike the grid, the speedup on Chicago Sketch is *not* monotone in richness: it peaks near $\bar{K} \approx 3.5$ and settles close to unity beyond $\bar{K} \approx 9$, while the feasible gap grows from 0.03% to 1.88%. Equation (11) explains the difference between the two families. Chicago Sketch paths average about 15 links, so even at $r = 20$ the dense term $2r = 40$ exceeds the sparse term; grid paths at $N = 32$ average 35–45 links, and the inequality runs the other way. The compression is thus best suited to networks with long paths and many parallel routes, and the enriched-pool regime of a short-path network is its adverse case.

Applying the rank rule to the submitted threshold grid improves accuracy as well as time: at $r = 20$ the conversion δ_F vanishes at every threshold, the feasible objective gaps are 0.027%, 0.000%, 0.020% and 0.028% for $\tau = 0, 0.46, 1.06$ and 4.54, and the link-flow R^2 rises to between 0.9997 and 1.0000, compared with 0.990–0.996 at $r = 50$. The distribution of demand between major and minor paths at each threshold is reported in Appendix B.

[Note to co-authors: Chicago Regional and Philadelphia are not yet re-evaluated at the corrected rank. The submitted Regional pool (879,625 paths) is no longer on file; the closest available instance has 2,024,525 paths and 30 links per path, which by (11) should favour $r \approx 10$. A C++ rank-by-offset experiment on it is running. Referee 1 will reasonably note that our largest speedups come from Sioux Falls and a synthetic grid, so carrying at least one large road network at the corrected rank matters more than any other remaining item.]

5.9 Reading the Two Representational Choices Together

The subspace criterion and the affine offset act on different parts of the same representation, and both were measured by the same device: an instance solved twice, back to back, with one choice changed and everything else held fixed. Figure 3 shows every such pair as a displacement rather than as a point, which is the only form in which these comparisons are controlled. The absolute position of a configuration in the speed–accuracy plane is dominated by which instance it is; the displacement isolates the choice under study.

Two readings follow, and the noise floor decides which is which. Horizontally, the arrows in panel (a) point left in 27 of 34 pairs: the weighted subspace is an accuracy improvement, and it is measured exactly. Vertically they scatter, and only 2 of 34 exceed the noise floor, so the apparent speed gains are not resolvable and we do not claim them. Panel (b) is the offset ablation on the weighted representation, and the gap rises in 16 of 20 grid cells, confirming that removing the anchor loses accuracy whichever subspace is used.

Figure 4 shows the same two comparisons over the full grid design, on the axis that can be resolved. The weighted sheet lies below the plain one and the offset-free sheet lies above both, and the separation grows with pool depth, which is where the criterion has the most to gain.

5.10 Scope and Limitations

Four limitations bound the claims above, and each is a measurement rather than a caveat.

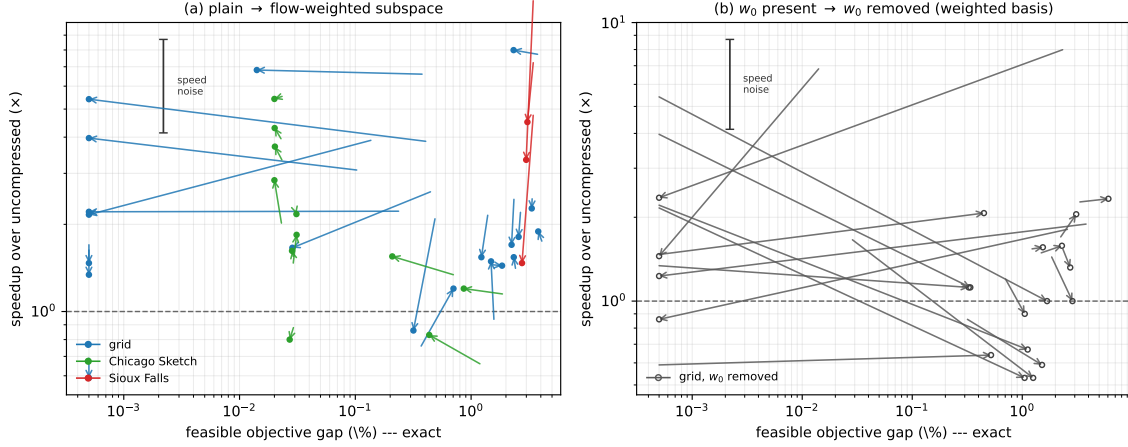


Figure 3: Paired comparisons. Each arrow is one instance solved twice with a single change: (a) plain to flow-weighted subspace, (b) affine offset present to removed. The horizontal axis is exact; the vertical axis carries the measurement noise of Section 5.1, drawn as the error bar, so vertical displacements shorter than it are not effects.

The speedup depends on the implementation. On the richness axis of Table 8 the Python implementation peaks at moderate richness and falls away, whereas the compiled implementation rises monotonically towards unity without crossing it; the compiled solver’s sparse kernels are relatively faster than its dense ones, which moves the balance in (11). Sioux Falls is favourable in both (3.6 $\times$ and 6.2 $\times$). Speedup is therefore a property of representation, operator, implementation and instance geometry jointly, and we report it only within an implementation.

The dense reconstruction is the binding constraint. Reducing the number of independent variables does not reduce the $n - s$ reconstructed inequalities. This is what (11) formalizes, what the falling branch of Table 8 measures, and what limits the method on short-path networks with rich pools.

A grouped representation removes that limit but is not yet faster. Replacing the signed subspace by nonnegative groups makes feasibility a bound constraint by construction, with no reconstruction at all. Measured on the adverse cells of Table 8, it is essentially exact — feasible gaps of 0.016% and 0.000%, in one case improving on the uncompressed solution — but in the present implementation it is not faster. It is the natural representation for the regime where the signed subspace struggles, and we leave its efficient implementation to future work.

Timing precision. Large-instance timings are single runs; repeated identical solves varied by up to 40% under machine load, so ratios near unity on those instances should not be read as ordered.

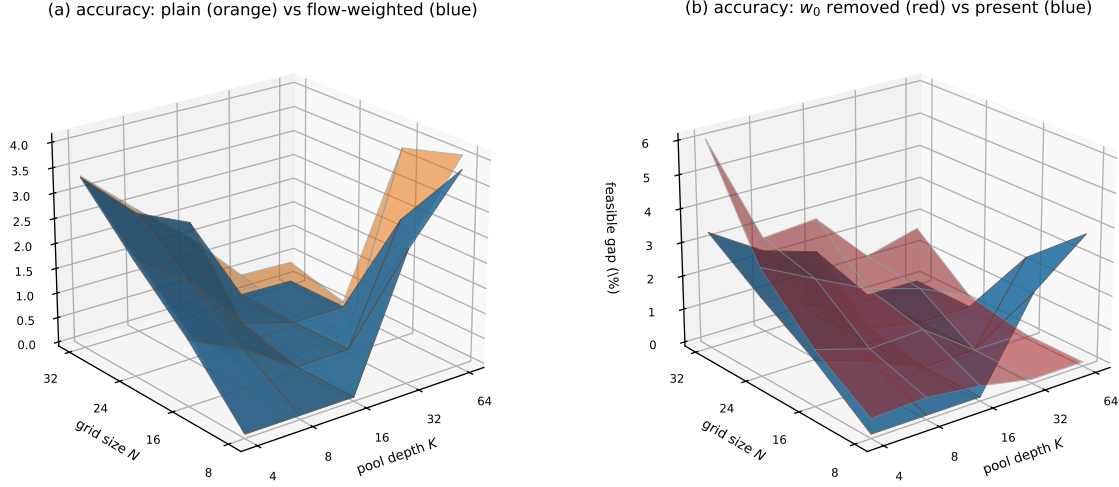


Figure 4: Feasible objective gap over the grid design (N, K) . (a) plain (orange) against flow-weighted (blue): the weighted sheet is lower in 15 of 20 cells. (b) offset present (blue) against removed (red): removing the anchor lifts the surface in 16 of 20 cells.

6 Conclusion and Discussion

We developed an integrated compressed traffic assignment framework in which nominal path flows identify major paths, deviations of the minor-path flows are represented through the fixed map $w = w_0 + \Phi_r z$, and the SVD choice $\Phi_r = U_r$ supplies the approximation used by the augmented Lagrangian implementation. The contribution is this unified compression-and-solution construction rather than the SVD or augmented Lagrangian principle in isolation.

The computational study is organized around a cost model rather than a parameter sweep. Because the reconstruction $w_0 + U_r z \geq 0$ is dense, a compressed gradient pays about $2r$ operations per minor path where the uncompressed gradient pays the length of a path, so compression reduces work when $2r$ is at most the average path length. A controlled grid family confirms the prediction, with speedups rising with candidate-pool depth to *6.8 times* at unchanged accuracy; a rank sweep shows that over the useful range the rank is almost purely a speed parameter, changing solution time by an order of magnitude while the feasible objective gap moves in the fourth decimal place; choosing the retained subspace by flow-weighted rather than plain reconstruction error reduces the feasible objective gap by more than half on route-rich instances at no additional preprocessing cost, an accuracy gain that the measurement precision resolves cleanly while the accompanying time differences do not; and an exact ablation shows that the affine offset is load-bearing, since a signed subspace of nonnegative flows is usable only relative to an interior anchor. These last two

act on different parts of the representation and their benefits do not overlap. Comparing operators on the same compressed instances shows that the benefit is specific to the augmented Lagrangian: variable reduction accelerates a method whose per-iteration bottleneck lies in the variables, whereas projection- and reduced-gradient-type methods depend on a separable feasible set that compression destroys. This is the measured justification for the pairing developed in this paper.

The scope remains conservative. The reconstructed nonnegativity system contains path-scale components and U_r is generally dense. The factored and mixed implementations show that this density did not eliminate per-inner-iteration savings in the tested regime, but total-time gains remain problem-dependent because compression can increase the iteration count. The framework is most appropriate when a path pool and nominal flow are available and related assignments allow the compression factors to be reused. Extensions requiring a different representation, adaptive path management, or other network-optimization domains are outside the scope of this paper.

References

- [AMO93] Ravindra K. Ahuja, Thomas L. Magnanti, and James B. Orlin. *Network Flows: Theory, Algorithms, and Applications*. Prentice Hall, Englewood Cliffs, NJ, 1993.
- [Ber80] Dimitri P. Bertsekas. A class of optimal routing algorithms for communication networks. Technical Report LIDS-P-1042, Massachusetts Institute of Technology, Laboratory for Information and Decision Systems, Cambridge, MA, 1980. Presented at ICC 80, Atlanta, GA, Oct. 1980.
- [Ber82] Dimitri P. Bertsekas. *Constrained Optimization and Lagrange Multiplier Methods*. Academic Press, 1982. Republished by Athena Scientific, Belmont, MA, 1997.
- [Ber98] Dimitri P. Bertsekas. *Network Optimization: Continuous and Discrete Models*. Athena Scientific, Belmont, MA, 1998.
- [Ber09] Dimitri P. Bertsekas. *Convex Optimization Theory*. Athena Scientific, Belmont, MA, 2009.
- [Ber11] Dimitri P. Bertsekas. Centralized and distributed newton methods for network optimization and extensions. Technical Report LIDS-2866, Massachusetts Institute of Technology, Laboratory for Information and Decision Systems, Cambridge, MA, April 2011.
- [Ber12] Dimitri P. Bertsekas. *Dynamic Programming and Optimal Control*, volume 2. Athena Scientific, 4 edition, 2012.

- [Ber16] Dimitri P. Bertsekas. *Nonlinear Programming*. Athena Scientific, 3 edition, 2016.
- [Ber19] Dimitri P. Bertsekas. *Reinforcement Learning and Optimal Control*. Athena Scientific Belmont, MA, 2019.
- [BG02] Hillel Bar-Gera. Origin-based algorithm for the traffic assignment problem. *Transportation Science*, 36(4):398–417, 2002.
- [BGG84] Dimitri Bertsekas, Eli Gafni, and Robert Gallager. Second derivative algorithms for minimum delay distributed routing in networks. *IEEE Transactions on Communications*, 32(8):911–919, 1984.
- [BJN⁺98] Cynthia Barnhart, Ellis L. Johnson, George L. Nemhauser, Martin W.P. Savelsbergh, and Pamela H. Vance. Branch-and-price: Column generation for solving huge integer programs. *Operations Research*, 46(3):316–329, 1998.
- [BLU20] Stephen D. Boyles, Nicholas E. Lownes, and Avinash Unnikrishnan. Transportation network analysis. *Volume I: Static and Dynamic Traffic Assignment*, 2020.
- [BMW56] Martin Beckmann, C. McGuire, and C. Winsten. *Studies in the economics of transportation*. Yale University Press, 1956.
- [Cam15] Pedro Camargo. Aequilibræ: a free qgis add-on for transportation modeling. *Foss4g North America*, 2015.
- [CR09] Emmanuel J. Candès and Benjamin Recht. Exact matrix completion via convex optimization. *Foundations of Computational Mathematics*, 9(6):717–772, 2009.
- [CZ93] Vijay Kumar Chopra and William T. Ziemba. The effect of errors in means, variances, and covariances on optimal portfolio choice. *Journal of Portfolio Management*, 19(2):6–11, 1993.
- [EY36] Carl Eckart and Gale Young. The approximation of one matrix by another of lower rank. *Psychometrika*, 1(3):211–218, 1936.
- [FW56] Marguerite Frank and Philip Wolfe. An algorithm for quadratic programming. *Naval Research Logistics Quarterly*, 3(1–2):95–110, 1956.
- [Gal77] Robert Gallager. A minimum delay routing algorithm using distributed computation. *IEEE Transactions on Communications*, 25(1):73–85, 1977.
- [GK65] Gene H. Golub and William Kahan. Calculating the singular values and pseudo-inverse of a matrix. *Journal of the Society for Industrial and Applied Mathematics, Series B: Numerical Analysis*, 2(2):205–224, 1965.

- [GVL13] Gene H. Golub and Charles F. Van Loan. *Matrix Computations*. Johns Hopkins University Press, 4 edition, 2013.
- [HPC09] Shou-Ren Hu, Srinivas Peeta, and Chun-Hsiao Chu. Identification of vehicle sensor locations for link-based network traffic applications. *Transportation Research Part B: Methodological*, 43(8-9):873–894, 2009.
- [JTPR94] R. Jayakrishnan, W. K. Tsai, J. Prashker, and S. Rajadhyaksha. A faster path-based algorithm for traffic assignment. *Transportation Research Record*, 1443:75–83, 1994.
- [KBV09] Yehuda Koren, Robert Bell, and Chris Volinsky. Matrix factorization techniques for recommender systems. *Computer*, 42(8):30–37, 2009.
- [LCH⁺23] Zhiyuan Liu, Xinyuan Chen, Jintao Hu, Shuaian Wang, Kai Zhang, and Honggang Zhang. An alternating direction method of multipliers for solving user equilibrium problem. *European Journal of Operational Research*, 310(3):1072–1084, 2023.
- [LD05] Marco E. Lübbecke and Jacques Desrosiers. Selected topics in column generation. *Operations Research*, 53(6):1007–1023, 2005.
- [LDZ⁺24] Zhiyuan Liu, Yu Dong, Honggang Zhang, Nan Zheng, and Kai Huang. A novel parallel computing framework for traffic assignment problem. *Transportation Research Part E: Logistics and Transportation Review*, 189:103687, 2024.
- [LSX⁺25] Pengjie Liu, Hu Shao, Shengbei Xu, Emily Zhu Fainman, and Chunkai Tang. An ADMM-based splitting algorithm with multiplier sequential updates for solving traffic assignment. *Transportmetrica A: Transport Science*, pages 1–33, 2025.
- [LZ25a] Peiheng Li and Xuesong Zhou. OpenDTA, 2025. Retrieved from <https://github.com/jdlph/OpenDTA>.
- [LZ25b] Peiheng Li and Xuesong Zhou. Path4GMNS, 2025. Retrieved from <https://github.com/jdlph/Path4GMNS>.
- [Mar52] Harry Markowitz. Portfolio selection. *The Journal of Finance*, 7(1):77–91, 1952.
- [ML13] Maria Mitradjieva and Per Olov Lindberg. The stiff is moving—Conjugate direction Frank-Wolfe methods with applications to traffic assignment. *Transportation Science*, 47(2):280–293, 2013.
- [Nie10] Yu Marco Nie. A class of bush-based algorithms for the traffic assignment problem. *Transportation Research Part B: Methodological*, 44(1):73–89, 2010.

- [Pat94] Michael Patriksson. *The Traffic Assignment Problem: Models and Methods*. VSP, Utrecht, The Netherlands, 1994.
- [PB99] Srinivas Peeta and Srinivas Bulusu. Generalized singular value decomposition approach for consistent on-line dynamic traffic assignment. *Transportation Research Record*, 1667(1):77–87, 1999.
- [PSA⁺17] A. Arun Prakash, Ravi Seshadri, Constantinos Antoniou, Francisco C. Pereira, and Moshe E. Ben-Akiva. Reducing the dimension of online calibration in dynamic traffic assignment systems. *Transportation Research Record*, 2667(1):96–107, 2017.
- [Ros76] Stephen A. Ross. The arbitrage theory of capital asset pricing. *Journal of Economic Theory*, 13(3):341–360, 1976.
- [RR08] Ali Rahimi and Benjamin Recht. Random features for large-scale kernel machines. In *Advances in Neural Information Processing Systems*, volume 20, pages 1177–1184, 2008.
- [She85] Yosef Sheffi. *Urban Transportation Networks: Equilibrium Analysis with Mathematical Programming Methods*. Prentice-Hall, Englewood Cliffs, NJ, 1985.
- [SK87] Lawrence Sirovich and Michael Kirby. Low-dimensional procedure for the characterization of human faces. *Journal of the Optical Society of America A*, 4(3):519–524, 1987.
- [SKKR00] Badrul Sarwar, George Karypis, Joseph Konstan, and John Riedl. Application of dimensionality reduction in recommender system—a case study. In *Proceedings of the ACM WebKDD Workshop*, pages 82–90, 2000.
- [SSM98] Bernhard Schölkopf, Alexander Smola, and Klaus-Robert Müller. Nonlinear component analysis as a kernel eigenvalue problem. *Neural Computation*, 10(5):1299–1319, 1998.
- [Vap95] Vladimir N. Vapnik. *The Nature of Statistical Learning Theory*. Springer, 1995.
- [XNL18] Jun Xie, Yu Nie, and Xiaobo Liu. A greedy path-based algorithm for traffic assignment. *Transportation Research Record*, 2672(48):36–44, 2018.
- [YZD⁺19] Yu Yao, Xiaoning Zhu, Hongyu Dong, Shengnan Wu, Hailong Wu, Lu Carol Tong, and Xuesong Zhou. ADMM-based problem decomposition scheme for vehicle routing problem with time windows. *Transportation Research Part B: Methodological*, 129:156–174, 2019.

- [ZCW⁺22] Xuesong Zhou, Qixiu Cheng, Xin Wu, Peiheng Li, Baloka Belezamo, Jiawei Lu, and Mohammad Abbasi. A meso-to-macro cross-resolution performance approach for connecting polynomial arrival queue model to volume-delay function with inflow demand-to-capacity ratio. *Multimodal Transportation*, 1(2):100017, 2022.
- [ZL10] Xuesong Zhou and George F. List. An information-theoretic sensor location model for traffic origin-destination demand estimation applications. *Transportation Science*, 44(2):254–273, 2010.
- [ZQ25] Pengji Zhang and Sean Qian. Low-rank approximation of path-based traffic network models. *Transportation Research Part C*, 150:104151, 2025.
- [ZT14] Xuesong Zhou and Jeffrey Taylor. Dtalite: A queue-based mesoscopic traffic simulator for fast model evaluation and calibration, 2014.

A Notation

Table 9: Summary of Main Notation

Symbol	Description
Network and Demand	
d	OD demand vector (multi-path ODs only)
v	Link flow vector
v_0	Link flow from singleton paths
m	Number of links
n	Number of paths (excluding singletons)
s	Number of major paths
ℓ	Number of multi-path OD pairs
Path Variables	
x	Full path flow vector
y	Major-path flow vector
w	Minor-path flow vector
w_0	Nominal minor-path flow vector
z	Low-dimensional minor-path coordinate vector
Incidence Matrices	
B	Path-link incidence matrix ($n \times m$)
B_1	Incidence matrix for major paths ($s \times m$)
B_2	Incidence matrix for minor paths $[(n - s) \times m]$

Symbol	Description
A	OD-path incidence matrix ($\ell \times n$)
A_1	OD-major-path incidence matrix ($\ell \times s$)
A_2	OD-minor-path incidence matrix [$\ell \times (n - s)$]
$D = B'_2 U_r$	Compressed link matrix ($m \times r$)
$M = A_2 U_r$	Compressed OD matrix ($\ell \times r$)
Low-Rank Compression	
$B_2 \approx U_r \Sigma_r V'_r$	Rank- r truncated SVD of minor-path matrix
U_r	Rank- r left singular basis for minor paths
r	Compression rank
$w_0 + U_r z$	Affine compressed minor-path flow representation
$\bar{d} = d - A_2 w_0$	OD demand remaining after the nominal minor-path contribution
Optimization and ALM	
$\hat{f}(y, z) = f(B'_1 y + Dz + B'_2 w_0 + v_0)$	Compressed objective function
λ	Lagrange multiplier vector (OD equality constraints)
μ	Lagrange multiplier vector (nonnegativity of $w_0 + U_r z$)
$c = (c_1, c_2)$	Component-wise penalty parameters
β	Penalty growth factor
γ	Penalty update threshold

B Path and Flow Distribution Tables

This appendix reports how demand splits between major and minor paths at each cutoff threshold. The three networks display markedly different concentration profiles.

On Chicago Sketch (Table 10), major paths carry roughly 23% of total demand across all thresholds and minor-path flow stays below 1.5% even at the most aggressive cutoff, indicating that the vast majority of demand is absorbed by singleton OD pairs that lie outside the compressed problem. On Chicago Regional (Table 11), major paths carry 47–59% of demand, with minor-path flow rising to 11.7% at $\tau = 1.033$; the heavier tail makes this network the most sensitive to the choice of τ . On Philadelphia (Table 12), major paths dominate at 63–71% of demand and minor-path flow reaches 7.8% at $\tau = 5.3344$. In all three networks, the fraction of flow routed through minor paths grows smoothly with τ , confirming that the cutoff acts as a continuous control over the major/minor partition rather than a sharp threshold.

Table 10: Path and Flow Distribution (Chicago Sketch)

τ (veh)	s (major)	$n - s$ (minor)	Major Flow (%)	Minor Flow (%)
0.000	42,774	0	267,805.15 (23.5)	0.00 (0.0)
0.139	40,243	2,531	267,509.30 (23.5)	295.85 (0.0)
0.341	32,650	10,124	265,813.58 (23.4)	1,991.57 (0.2)
1.055	25,057	17,717	261,257.02 (23.0)	6,548.13 (0.6)
4.541	19,995	22,779	250,289.70 (22.0)	17,515.45 (1.5)

Table 11: Path and Flow Distribution (Chicago Regional)

τ (veh)	s (major)	$n - s$ (minor)	Major Flow (%)	Minor Flow (%)
0.000	879,625	0	778,822.96 (59.2)	0.00 (0.0)
0.126	827,019	52,606	772,855.35 (58.7)	5,967.60 (0.5)
0.227	669,208	210,417	745,814.50 (56.7)	33,008.46 (2.5)
0.460	511,394	368,231	694,810.96 (52.8)	84,011.99 (6.4)
1.033	406,187	473,438	624,290.35 (47.4)	154,532.61 (11.7)

Table 12: Path and Flow Distribution (Philadelphia)

τ (veh)	s (major)	$n - s$ (minor)	Major Flow (%)	Minor Flow (%)
0.0000	1,846,068	0	10,157,630.99 (70.9)	0.00 (0.0)
0.1610	1,721,624	124,444	10,141,117.22 (70.8)	16,513.78 (0.1)
0.4473	1,348,298	497,770	10,037,598.50 (70.1)	120,032.49 (0.8)
0.9866	1,099,414	746,654	9,867,685.32 (68.9)	289,945.67 (2.0)
2.4435	850,528	995,540	9,481,879.91 (66.2)	675,751.08 (4.7)
5.3344	726,087	1,119,981	9,036,971.50 (63.1)	1,120,659.49 (7.8)